Given:- $L_1 = 0.04H$. $L_2 = 0.2H$

$R_1 = 25\Omega$ $R_2 = 15\Omega$.

$V = 230V$, $F = 50Hz$.

Q2.

To find:- $P_{d1} = ?$ $PF_1 = ?$

$P_{d2} = ?$ $PF_2 = ?$.

Solution:- $X_{L1} = 2\pi f L_1 = 2\pi \times 50 \times 0.04$

$= 12.57\ \Omega$

$X_{L2} = 2\pi f L_2 = 2\pi \times 50 \times 0.2$

$= 62.83\ \Omega$

Total resistance $= R_1 + R_2 = 40\Omega$

Total inductive reactance $= X_{L1} + X_{L2}$

$= 75.4\ \Omega$

Impedance $Z = (40 + j\,75.4)\Omega$

$= (85.35 < 62.05)\Omega$

Current $\bar{I} = \dfrac{\bar{V}}{\bar{Z}} = \dfrac{230 < 0^\circ}{85.35 < 62.05}$

$I = 2.695 < -62.05$

p.d in coil 1 $= P_1 = I^2 \times R_1$

$= (2.695)^2 \times 25$

$P_1 = 181.57$ Watts.

p.d in coil 2 $= I^2 \times R_2 = (2.695)^2 \times 15$

$P_2 = 108.95$ Watts

Phase angle $\phi = 62.05$

p.f $= \cos\phi = \cos 62.05 = 0.4687$ (Lagging)

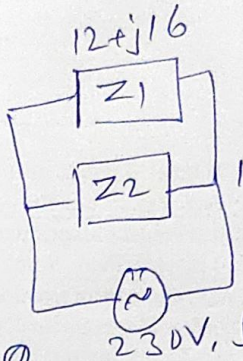
Q.5

Given :- $Z_1 = 12 + j16\ \Omega = 20 \angle 53.13$

$Z_2 = 10 - j20\ \Omega = 22.36 \angle -63.43$

$V_L = 230\ V$

$f = 50\ Hz$.



To find :-

Admittance $Y = ?$

$I = ?$

PF of each branch $= ?$

Solution :- $Z = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{(20 \angle 53.13)(22.36 \angle -63.43)}{(12 + j16) + (10 - j20)}$

$= \frac{447.2 \angle -10.3}{22 - j4}$

$= \frac{447.2 \angle -10.3}{22.36 \angle -10.3}$

$= 20 \angle 0^\circ$

$Y = \frac{1}{Z} = \frac{1}{20 \angle 0^\circ} = \frac{1 \angle 0^\circ}{20 \angle 0^\circ}$

$= 0.05 \angle 0^\circ$

$= 0.05 + j0\ \Omega$.

$I = \frac{V}{Z} = \frac{230 \angle 0^\circ}{20 \angle 0^\circ} = 11.5 \angle 0^\circ\ A$

$PF_1 = \cos \phi_1 = \cos 53.13^\circ = 0.6$ (lagging)

$PF_2 = \cos \phi_2 = \cos(-63.43) = 0.447$. (leading)

Q6. A balanced 3-ϕ delta connected load with per phase impedance of 20 Ω. Determine line current and phase current. Given $V_L = 1000$ V.

Given :- $Z_{ph} = 20\,\Omega$, $V_L = 1000$ V

Soln :- For delta connection -

$$V_{ph} = V_L = 1000\text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{1000}{20} = 5\text{ A}$$

$$I_L = \sqrt{3}\, I_{ph} = \sqrt{3} \times 5 = 8.66\text{ A}$$

Q1 :- (Star connected)

Given :- $R = 6\,\Omega$, $X_L = 8\,\Omega$ (3 coils)

$V = 400$ V $= V_{ph}$.

$f = 50$ Hz.

To find :- $I_L = ?$

Solution :- $Z_{ph} = R + jX_L = 6 + j8$

$= 10 \angle 53.13$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94\text{ V}.$$

$$I_L = I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{10} = 23.09\text{ A}$$

$= I_L$.

$PF = \cos\phi = \cos 53.13$

$= 0.6$ (lagging)